Huawei Access Node BTM-ACC-23 Hardware/Environmental Fault Troubleshooting SOP

Node: BTM-ACC-23 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2790

Description

This Standard Operating Procedure outlines the troubleshooting and resolution steps for hardware/environmental faults occurring on the Access Node BTM-ACC-23, running under Huawei's CLI dialect. The failure is attributed to ASIC memory parity errors, line card crashes, or power supply redundancy loss, leading to the critical alarm code ALARM_2790. The underlying cause can be attributed to the high volume of traffic being processed by the Access Node, resulting in excessive heat generation and stress on the hardware components. Furthermore, the use of Huawei's advanced ASIC technology, which integrates multiple functions onto a single chip, can lead to complex fault scenarios. As a result, the failure of a single component can have a cascading effect on the entire system, requiring a methodical and structured approach to diagnosis and resolution.

Pre-requisite Checks

- Verify that the site location, BTM Layout Aggregation, has been dispatched to and has acknowledged receipt of the incident.

- Check the status of the redundant links and confirm that the system is not in a split-configuration state.

- Ensure that all necessary personnel and equipment are on site and ready to assist with the troubleshooting and repair process.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: display version, Expected Output: System version information, including the hardware revision level and software image version., Description: This command will provide information about the system version, including the hardware revision level and software image version, which can help identify potential issues related to ASIC memory parity errors or line card crashes.

Command: display board, Expected Output: Detailed information about the line cards, including status and temperature information., Description: This command will provide detailed information about the
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line cards, including status and temperature information, which can help identify potential issues related to line card crashes or excessive heat generation.

Command: display power supply, Expected Output: Information about the power supply status, including redundancy and voltage levels., Description: This command will provide information about the power supply status, including redundancy and voltage levels, which can help identify potential issues related to power supply redundancy loss.
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Resolution Steps

Step 1: Step: Replace the faulty line card, Command: replace line-card , Description: This command will initiate the replacement of the faulty line card with a new one, which should resolve the issue related to line card crashes.

Step 2: Step: Replace the faulty ASIC, Command: replace asic , Description: This command will initiate the replacement of the faulty ASIC with a new one, which should resolve the issue related to ASIC memory parity errors.

Step 3: Step: Replace the power supply unit, Command: replace power-supply , Description: This command will initiate the replacement of the faulty power supply unit with a new one, which should resolve the issue related to power supply redundancy loss.

Post-Check Verification

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Command: display alarm, Expected Output: No alarm information or a cleared alarm status., Description: This command will verify that the alarm code ALARM_2790 has been cleared.

Command: display board, Expected Output: Normal line card status and temperature information., Description: This command will verify that the line card status and temperature information are normal.

Command: display power supply, Expected Output: Normal power supply status, including redundancy and voltage levels., Description: This command will verify that the power supply status is normal.
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Roll-back Procedures

- Step: Revert the replacement of the faulty component, Command: revert replace , Description: This command will revert the replacement of the faulty component with the original faulty one, which can help restore the system to its previous state.

- Step: Revert the configuration changes, Command: revert config, Description: This command will revert the configuration changes made during the troubleshooting process, which can help restore the system to its previous state.